1.Project overview :

Quick pay is an application used in transferring and receiving money . It allows users to transfer and receive money

For admins , there are other features such as knowing everything about users and their transactions

2.Functional requirements :

Functions for user:

"""""""""""""""""""""""""""""

1- transfer money

2- add favorites

3-remove favorites

4-show favorites

5- show balances

6- show transaction history

7- search recipient

8- update profile

9- view profile

Functions for admin:

"""""""""""""""""""""""""""""""""""

1- Show Statistics

2- Search User

3- View All Users

4- Update User

5- View All Transactions

6- Block Account

7- Unblock Account

8- View User Details

3.Business Rules:

Transferring: users send money to others in case some conditions are valid such as both accounts of sender and receiver are active , transaction history isn't full and amount of money is valid.

Balances: balances are added to the application in case :they are not less than zero . Users can't send amounts of money greater than their balances .

Account Blocking: blocking users mean users are not allowed to send money, receive money or do any actions on the application

User permissions: users are allowed to send money , receive money , add favourites , search a user or do anything except for actions done by admins such as (un)blocking users , show statistics , view all users or update users .

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4. Data Structure Explanation:

There are totally 16 arrays in the project :

"FOR USERS"

1) 1d array for users' names : to store all names

2) 1d array for users' ID : to store all IDs

3) 1d array for users' email addresses : to store all emails of users

4) 1d array for users' phone numbers : to store all phone numbers of users

5) 1d array for users' passwords : to store all passwords of users

6) 1d array for balances of users : to store all balances of users

7) 1d array for statuses of users (blocked or not) : to store status of users whether they are blocked or not

"FOR ADMIN":

1) 1d array for admins' passwords : to store passwords of all admins

2) 1d array for admins' emails : to store emails of all admins

"FOR TRANSACTION":

1) 1d array for transaction IDs : to store id of all transactions

2) 1d array for transaction senders: to store names of transactions senders

3) 1d array for transaction receivers: to store names of receivers in transaction process

4) 1d array for transaction amounts : to store amounts sent or received during transactions

5) 1d array for transaction statuses : to store statuses of transaction (successful or not)

"FOR FAVOURITES"

1) 2d array for favourite contacts : to store the favourites of each user

2) 1d array for favourite counts : to store the number of favourite contacts for users.

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5. Methods Documentation :

USER METHODS :

"""""""""""""""""""""""""""""""

1. Show Balance : The showBalance(int index) method displays the balance of the logged-in user. It uses the user's index to get the correct value from the Balances array.

2. Transfer Money : The transferMoney(int index) method allows a user to send money to another user. It checks that the sender is active, finds the recipient, checks that the recipient is active, makes sure the amount is positive, and checks that the sender has enough money. If everything is valid, it updates both balances and saves the transaction details in the transaction arrays.

3. Transaction History : The showTransactionHistory(int index) method displays transactions connected to the logged-in user. It shows the transaction ID, sender, receiver, amount, and status. It uses a temporary two-dimensional array to organize the information before printing it.

4. Show Favorites : allows users to see their list of favourite contacts

5. Add Favorite : allows users to add contacts to favourite list to get to them easily while using application

6. Remove Favorite: allows users to remove users from favourite contacts

7. Search Recipient Method :

This method is used to search for a recipient before transferring money to make sure that the recipient exists and is the correct person the user wants to send money to.

The user can search for the recipient using their email address or phone number. If the recipient is found, the system displays their name, email address, and phone number so the user can confirm their identity before making the transfer.

For privacy and security reasons, the method must not display sensitive information such as the recipient's balance, account status, or any other private information.

If the recipient does not exist, the system displays a message informing the user that the recipient was not found.

8. View Profile: allows the user to see account details

9. Update Profile : Allows the user to update their account, subject to password validation.

ADMIN METHODS :

"""""""""""""""""""""""""""""""

1. Show Statistics : The showStatistics() method displays general information about the system. It counts active and blocked users, calculates the total balance, and shows transaction statistics such as successful transfers and the total amount transferred.

2. Search User: Ask the user for the search method, perform the search, and then retrieve the remaining data using the index.

3. View All Users : Prints all names, IDs, and emails of the users.

4. Update User : Allows the admin to modify user accounts.

5. Admin Transaction History : The adminShowTransactionHistory() method asks the admin to enter a user's ID. It finds that user and displays transactions where the user is the sender or the receiver.

6. (Un)block Account : Allows the admin to block or unblock users.

8. View User Details : Allows an admin to search for a user and view their account details.

6. Test cases :

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1) Successful money transfer :

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Sender balance = 16000

Amount to transfer = 2000

Receiver balance = 18000

Receiver balance becomes = 20000

Sender balance becomes = 14000

Result : transfer successful

2) Transfer with insufficient balance :

Amount to transfer = 20000

Sender balance = 16000

Receiver balance = 18000

Result : insufficient balance

3)Transfer to a nonexistent user :

Result : recipient not found

4) Transfer to blocked account:

Result : your account is blocked

5) Transfer of zero or negative amount:

Result : amount must be greater than zero

6) Search for existent or non existent user:

Result (existent) :

recipient found

Name : ---

Email : ---

Phone number : ---

Result (non existent): recipient not found

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7. How to run application :

First : login as user or admin or exit

Second : if user , enter your email , password and then user panel is displayed to choose a function

Third : if admin , enter your email , password and then admin panel is displayed to choose a function